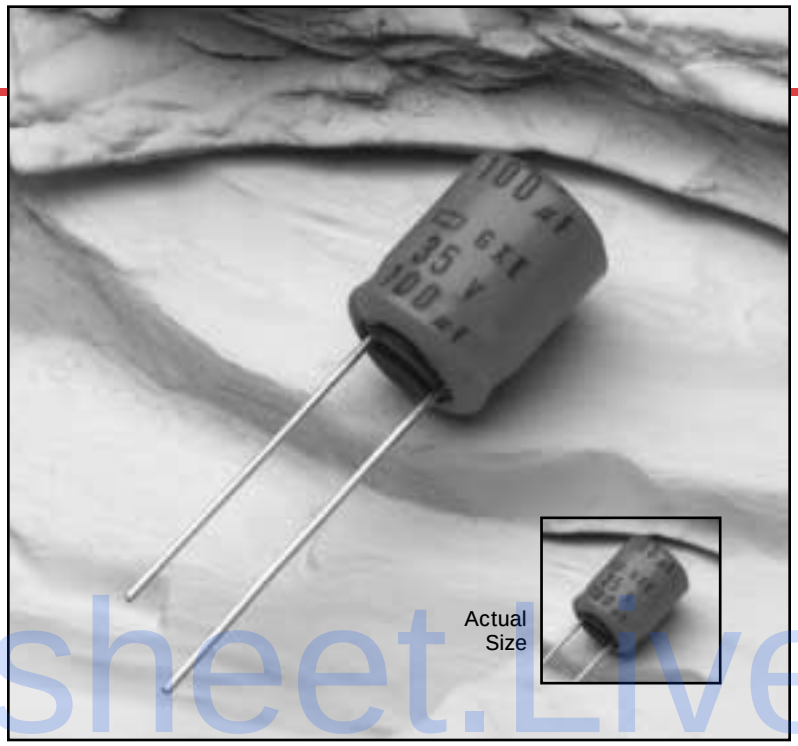
GXE Series

UNITED CHEMI-CON

- Miniature
- Long Life
- High CV
- High Reliability
- +125°C Maximum Temperature



Datasheet.Live

The GXE series capacitors are new miniature high temperature capacitors from United Chemi-Con. These capacitors are designed for use in very high temperature applications such as automotive engine controls. These GXE capacitors are an improved series offering longer useful lifetimes when compared to the older GXD series.

The GXE series capacitors are non-solvent proof. Refer to the Mini-Glossary for cleaning guidelines and recommended cleaning agents that are compatible with United Chemi-Con products.

Summary of Specifications

- **Radial lead terminals.**
- **Capacitance range: 4.7 to 4,700µF.**
- **Voltage range: 10 to 450VDC.**
- **Category temperature range: −40°C to +125°C for 10 to 250V; −25°C to +125°C for 350 to 450V.**
- **Leakage current: See specifications table for leakage current at +20°C.**
- **Standard capacitance tolerance: ±20%**
- **Nominal case size (D×L): 8×12mm to 16×31.5mm.**
- **Rated lifetime: 2,000 to 5,000 hours at +125°C with the rated ripple current applied, depending on case size.**

GXE Series

GXE Specifications

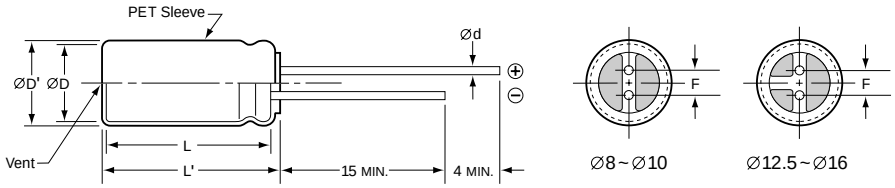
<table>
<tr><th>Item</th><th>Characteristics</th></tr>
<tr><td>Category Temperature Range</td><td>−40 to +125°C for 10 to 250VDC; −25 to +125°C for 350 to 450VDC</td></tr>
<tr><td>Rated Voltage Range</td><td>10 to 450VDC</td></tr>
<tr><td>Capacitance Range</td><td>4.7 to 4,700μF</td></tr>
<tr><td>Capacitance Tolerance</td><td>±20% (M) at +20°C, 120Hz</td></tr>
<tr><td>Leakage Current</td><td>At +20°C
<table>
<tr><th>DC Rated Voltage</th><th>Test Time</th><th colspan="2">Leakage Current (μA)</th></tr>
<tr><td>10-100V</td><td>After 1 minute</td><td colspan="2">I = 0.03CV or 4μA, whichever is greater.</td></tr>
<tr><td>160-450V</td><td>After 1 minute</td><td>CV≤1,000: I = 0.1CV+40</td><td>CV>1,000: I = 0.04CV+100</td></tr>
</table>
Where I = Max. leakage current (μA), C = Nominal capacitance (μF) and V = Rated voltage (V)</td></tr>
<tr><td>Dissipation Factor (Tan δ)</td><td>At +20°C, 120Hz
<table>
<tr><td>Rated Voltage (V)</td><td>10</td><td>16</td><td>25</td><td>35</td><td>50</td><td>63</td><td>80</td><td>100</td><td>160-250</td><td>350-450</td></tr>
<tr><td>Tan δ (DF)</td><td>0.20</td><td>0.16</td><td>0.14</td><td>0.12</td><td>0.10</td><td>0.10</td><td>0.08</td><td>0.08</td><td>0.20</td><td>0.24</td></tr>
</table>
When nominal capacitance exceeds 1,000μF, add 0.02 to the values above for each 1,000μF increase.</td></tr>
<tr><td>Low Temperature Characteristics</td><td>At 120Hz, impedance (Z) ratio between the −25°C or −40°C value and +20°C value shall not exceed the values given below.
<table>
<tr><td>Rated Voltage (V)</td><td>10</td><td>16</td><td>25</td><td>35</td><td>50</td><td>63</td><td>80</td><td>100</td><td>160-250</td><td>350-450</td></tr>
<tr><td>Z(−25°C)/Z(+20°C)</td><td>3</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>3</td><td>6</td></tr>
<tr><td>Z(−40°C)/Z(+20°C)</td><td>6</td><td>4</td><td>4</td><td>4</td><td>4</td><td>4</td><td>4</td><td>4</td><td>6</td><td>–</td></tr>
</table></td></tr>
<tr><td>Rated Ripple Current Multipliers
Refer to Section 4 of the Mini-Glossary for explanation of Rated Ripple Current Multipliers.</td><td>Frequency (Hz) for 10-100V
<table>
<tr><th rowspan="2">Capacitance (μF)</th><th colspan="4">Frequency (Hz)</th></tr>
<tr><th>120</th><th>1k</th><th>10k</th><th>100k</th></tr>
<tr><td>4.7-100μF</td><td>0.40</td><td>0.75</td><td>0.90</td><td>1.00</td></tr>
<tr><td>220-470μF</td><td>0.50</td><td>0.85</td><td>0.94</td><td>1.00</td></tr>
<tr><td>1,000μF</td><td>0.60</td><td>0.87</td><td>0.95</td><td>1.00</td></tr>
<tr><td>2,200-3,300μF</td><td>0.75</td><td>0.90</td><td>0.95</td><td>1.00</td></tr>
<tr><td>4,700μF</td><td>0.85</td><td>0.95</td><td>0.98</td><td>1.00</td></tr>
</table>
Frequency (Hz) for 160-450V
<table>
<tr><th rowspan="2">Capacitance (μF)</th><th colspan="6">Frequency (Hz)</th></tr>
<tr><th>50</th><th>120</th><th>300</th><th>1k</th><th>10k</th><th>100k</th></tr>
<tr><td>4.7-33μF</td><td>0.75</td><td>1.00</td><td>1.25</td><td>1.50</td><td>1.75</td><td>1.80</td></tr>
<tr><td>47-150μF</td><td>0.80</td><td>1.00</td><td>1.15</td><td>1.30</td><td>1.40</td><td>1.50</td></tr>
</table></td></tr>
<tr><td>Endurance
(Load Life)</td><td>The following specifications shall be satisfied when the capacitors are restored to +20°C after subjecting them to DC voltage for the specified test time at +125°C with the rated ripple current applied. The sum of DC voltage and peak AC voltage must not exceed the full rated voltage of the capacitors.
<table>
<tr><th>Case Diameter</th><th>Test Time (10-100V)</th></tr>
<tr><td>ø8mm</td><td>2,000 Hours</td></tr>
<tr><td>ø10mm</td><td>3,000 Hours</td></tr>
<tr><td>ø12.5mm & Above</td><td>5,000 Hours</td></tr>
</table>
10-100V
Capacitance change:
≤ ±30% of initial measured value
Tan δ (DF):
≤ 300% of initial specified value
Leakage current:
≤ initial specified value
<table>
<tr><th>Case Diameter</th><th>Test Time (160-450V)</th></tr>
<tr><td>ø10mm & Above</td><td>2,000 Hours</td></tr>
</table>
160-450V
Capacitance change:
≤ ±20% of initial measured value
Tan δ (DF):
≤ 200% of initial specified value
Leakage current:
≤ initial specified value</td></tr>
<tr><td>Shelf Life</td><td>The following specifications shall be satisfied when the capacitors are restored to +20°C after exposing them for 1,000 hours (500 hours for ≥350V) at +125°C without voltage applied. The rated voltage shall be applied to the capacitors for a minimum of 30 minutes, at least 24 hours and not more than 48 hours before the measurements.
10-100V
Capacitance change:
≤ ±30% of initial measured value
Tan δ (DF):
≤ 300% of initial specified value
Leakage current:
≤ initial specified value
160-450V
Capacitance change:
≤ ±20% of initial measured value
Tan δ (DF):
≤ 200% of initial specified value
Leakage current:
≤ 500% of initial specified value</td></tr>
</table>

GXE Series

Diagram of Dimensions

VB / Radial Lead

Unit: mm



Gas escape end seal for all case diameters.

For optional lead configurations and tape and ammo packaging, refer to the beginning of the Miniature Section.

øD	øD' max.	L' max.	ød	F ± 0.5
8	øD + 0.5	L + 2.0	0.6	3.5
10	øD + 0.5	L + 2.0	0.6	5.0
12.5	øD + 0.5	L + 2.0	0.6	5.0
16	øD + 0.5	L + 2.0	0.8	7.5

Part Numbering System for GXE Series

When ordering, always specify complete catalog number for GXE Series.

GXE 35 VB 101 M 10X12 LL

- LL: Lead Length: LL is Standard.
- 10X12: Case Code: See Case Sizes in Tables.
- M: Capacitance Tolerance: M = ±20%
- 101: Capacitance Value: Expressed in Microfarads. The first two digits are significant figures, and the third digit indicates the number of zeros for capacitance of 100µF or more. R indicates the decimal point for capacitance less than 100µF (e.g. R10 = .10µF; 1R0 = 1.0µF; 10R = 10µF; 101 = 100µF; 102 = 1,000µF; 103 = 10,000µF).
- VB: Lead Configuration: VB = Radial Lead Terminals.
- 35: DC Rated Voltage: Expressed in Volts (e.g. 35 = 35WVDC).
- GXE: Series Name: Indicates Basic Capacitor Design.

Standard Voltage Ratings - VB / Radial Lead (10 - 100V)

Rated Voltage (WVDC)	Capacitance (µF)	Catalog Part Number	Nominal Case Size* D × L (mm)	Maximum Impedance (Ω) at +20°C, 100kHz	Rated Ripple Current (mA rms) at +125°C, 100kHz
10 Volts **13 Volts Surge**	220	GXE10VB221M8X12LL	8 × 12	0.320	340
	330	GXE10VB331M10X12LL	10 × 12.5	0.150	620
	470	GXE10VB471M10X12LL	10 × 12.5	0.150	620
	1,000	GXE10VB102M10X20LL	10 × 20	0.075	950
	2,200	GXE10VB222M12X25LL	12.5 × 25	0.040	1,350
	3,300	GXE10VB332M16X25LL	16 × 25	0.031	1,620
	4,700	GXE10VB472M16X31LL	16 × 31.5	0.025	1,860

***The case sizes in table are with no sleeve, refer to diagram for case sizes with sleeve.**

GXE Series

Standard Voltage Ratings - VB/Radial Lead (10-100V)

Rated Voltage (WVDC)	Capacitance (µF)	Catalog Part Number	Nominal Case Size* D×L (mm)	Maximum Impedance (Ω) at +20°C, 100kHz	Rated Ripple Current (mA rms) at +125°C, 100kHz
16 Volts 20 Volts Surge	100	GXE16VB101M8X12LL	8 × 12	0.320	340
	220	GXE16VB221M10X12LL	10 × 12.5	0.150	620
	330	GXE16VB331M10X12LL	10 × 12.5	0.150	620
	470	GXE16VB471M10X16LL	10 × 16	0.094	790
	1,000	GXE16VB102M12X20LL	12.5 × 20	0.058	1,080
	2,200	GXE16VB222M16X25LL	16 × 25	0.031	1,620
	3,300	GXE16VB332M16X31LL	16 × 31.5	0.025	1,860
25 Volts 32 Volts Surge	100	GXE25VB101M8X12LL	8 × 12	0.320	340
	220	GXE25VB221M10X12LL	10 × 12.5	0.150	620
	330	GXE25VB331M10X16LL	10 × 16	0.094	790
	470	GXE25VB471M10X20LL	10 × 20	0.075	950
	1,000	GXE25VB102M12X25LL	12.5 × 25	0.040	1,350
	2,200	GXE25VB222M16X31LL	16 × 31.5	0.025	1,860
35 Volts 44 Volts Surge	100	GXE35VB101M8X12LL	8 × 12	0.320	340
	100	GXE35VB101M10X12LL	10 × 12.5	0.150	620
	220	GXE35VB221M10X16LL	10 × 16	0.094	790
	330	GXE35VB331M10X20LL	10 × 20	0.075	950
	470	GXE35VB471M12X20LL	12.5 × 20	0.058	1,080
	1,000	GXE35VB102M16X25LL	16 × 25	0.031	1,620
50 Volts 63 Volts Surge	10	GXE50VB10RM8X12LL	8 × 12	0.750	180
	22	GXE50VB22RM8X12LL	8 × 12	0.500	250
	33	GXE50VB33RM8X12LL	8 × 12	0.500	280
	47	GXE50VB47RM8X12LL	8 × 12	0.500	280
	100	GXE50VB101M10X12LL	10 × 12.5	0.200	520
	220	GXE50VB221M10X20LL	10 × 20	0.098	880
	330	GXE50VB331M12X20LL	12.5 × 20	0.081	990
	470	GXE50VB471M12X25LL	12.5 × 25	0.059	1,150
	1,000	GXE50VB102M16X31LL	16 × 31.5	0.032	1,590
63 Volts 79 Volts Surge	33	GXE63VB33RM8X12LL	8 × 12	1.500	150
	47	GXE63VB47RM10X12LL	10 × 12.5	0.590	530
	100	GXE63VB101M10X16LL	10 × 16	0.410	690
	220	GXE63VB221M12X20LL	12.5 × 20	0.160	1,050
	330	GXE63VB331M12X25LL	12.5 × 25	0.120	1,290
	470	GXE63VB471M12X30LL	12.5 × 30	0.097	1,460
	1,000	GXE63VB102M16X31LL	16 × 31.5	0.059	1,850
80 Volts 100 Volts Surge	22	GXE80VB22RM8X12LL	8 × 12	1.500	150
	33	GXE80VB33RM10X12LL	10 × 12.5	0.800	480
	47	GXE80VB47RM10X12LL	10 × 12.5	0.800	480
	100	GXE80VB101M10X20LL	10 × 20	0.390	790
	220	GXE80VB221M12X25LL	12.5 × 25	0.180	1,240
	330	GXE80VB331M12X30LL	12.5 × 30	0.160	1,390
	470	GXE80VB471M16X25LL	16 × 25	0.110	1,500
100 Volts 125 Volts Surge	4.7	GXE100VB4R7M8X12LL	8 × 12	2.000	130
	10	GXE100VB10RM8X12LL	8 × 12	1.500	150
	22	GXE100VB22RM10X12LL	10 × 12.5	0.800	480
	33	GXE100VB33RM10X12LL	10 × 12.5	0.800	480
	47	GXE100VB47RM10X16LL	10 × 16	0.550	630
	100	GXE100VB101M12X20LL	12.5 × 20	0.250	990
	220	GXE100VB221M16X25LL	16 × 25	0.110	1,500
	330	GXE100VB331M16X31LL	16 × 31.5	0.079	1,790

***The case sizes in table are with no sleeve, refer to diagram for case sizes with sleeve.**

GXE Series

Standard Voltage Ratings - VB/Radial Lead (160 - 450V)

Rated Voltage (WVDC)	Capacitance (µF)	Catalog Part Number	Nominal Case Size* D×L (mm)	Maximum Impedance (Ω) at +20°C, 100kHz	Rated Ripple Current (A rms) at +125°C, 120Hz
160 Volts **200 Volts Surge**	22	GXE160VB22RM10X20LL	10 × 20	4.000	115
	33	GXE160VB33RM10X25LL	10 × 25	2.600	154
	47	GXE160VB47RM12X20LL	12.5 × 20	1.800	187
	68	GXE160VB68RM12X25LL	12.5 × 25	1.300	245
	100	GXE160VB101M16X25LL	16 × 25	0.900	329
	150	GXE160VB151M16X31LL	16 × 31.5	0.590	434
200 Volts **250 Volts Surge**	10	GXE200VB10RM10X20LL	10 × 20	3.300	78
	22	GXE200VB22RM10X25LL	10 × 25	2.300	126
	33	GXE200VB33RM12X20LL	12.5 × 20	1.800	157
	47	GXE200VB47RM12X25LL	12.5 × 25	1.200	204
	68	GXE200VB68RM16X20LL	16 × 20	1.100	250
	100	GXE200VB101M16X25LL	16 × 25	0.740	329
250 Volts **300 Volts Surge**	10	GXE250VB10RM10X20LL	10 × 20	3.300	78
	22	GXE250VB22RM12X20LL	12.5 × 20	1.800	128
	33	GXE250VB33RM12X25LL	12.5 × 25	1.200	171
	47	GXE250VB47RM16X25LL	16 × 25	0.760	225
	68	GXE250VB68RM16X31LL	16 × 31.5	0.540	292
350 Volts **400 Volts Surge**	4.7	GXE350VB4R7M10X20LL	10 × 20	4.700	53
	10	GXE350VB10RM10X25LL	10 × 25	3.400	85
	22	GXE350VB22RM12X25LL	12.5 × 25	1.900	139
	33	GXE350VB33RM16X25LL	16 × 25	1.100	189
	47	GXE350VB47RM16X31LL	16 × 31.5	0.810	243
400 Volts **450 Volts Surge**	4.7	GXE400VB4R7M10X20LL	10 × 20	5.500	53
	10	GXE400VB10RM10X25LL	10 × 25	3.500	86
	22	GXE400VB22RM12X30LL	12.5 × 30	1.600	142
	33	GXE400VB33RM16X25LL	16 × 25	1.200	189
	47	GXE400VB47RM16X31LL	16 × 31.5	0.850	243
450 Volts **500 Volts Surge**	4.7	GXE450VB4R7M10X25LL	10 × 25	5.500	58
	10	GXE450VB10RM12X20LL	12.5 × 20	4.000	86
	22	GXE450VB22RM16X25LL	16 × 25	1.900	154
	33	GXE450VB33RM16X31LL	16 × 31.5	1.300	203

***The case sizes in table are with no sleeve, refer to diagram for case sizes with sleeve.**